

Cosmic Birefringence from Finite-Response Photon Transport in an Emergent Condensate Superfluid Medium

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Abstract

Recent analyses of cosmic microwave background (CMB) polarization data have reported hints of a small rotation of the linear polarization plane during photon propagation across cosmological distances, commonly referred to as *cosmic birefringence*. In the standard Λ CDM framework such a signal requires the introduction of new parity-violating fields or axion-like particles.

In this work we show that cosmic birefringence arises naturally in the Emergent Condensate Superfluid Medium (ECSM) framework. In ECSM, photon propagation occurs through a finite-response cosmological medium whose coherence state is characterized by a local variable χ . A small parity-odd component of the medium response produces a helicity-dependent phase shift between left- and right-circular polarization modes. This induces a rotation of the linear polarization field

$$Q + iU \rightarrow e^{2i\alpha}(Q + iU),$$

generating non-zero TB and EB correlations in the CMB polarization spectra.

Using the polarization transport formalism developed in the ECSM projection framework, we derive the expected rotation-induced mixing of E and B modes and compute the resulting TB and EB spectra. We show that a small rotation angle of order $\alpha \sim 0.1^\circ$ – 0.5° produces signals consistent with current observational sensitivities.

The ECSM mechanism predicts both an isotropic rotation component and a subdominant anisotropic contribution associated with coherence-gradient regions of the cosmological medium. These effects provide a direct observational probe of finite-response photon transport and offer a potential explanation of reported birefringence hints without introducing new fundamental particles.

1 Introduction

Measurements of the cosmic microwave background (CMB) polarization provide one of the most sensitive probes of photon propagation across cosmological distances. In the standard Λ CDM cosmological framework, the parity properties of Thomson scattering imply that the cross-correlation spectra TB and EB vanish in the absence of systematic effects.

However, several recent analyses of CMB polarization data have reported hints of a small rotation of the linear polarization plane during propagation from the surface of last scattering to the present epoch. This phenomenon, commonly referred to as *cosmic birefringence*, can be parameterized as a rotation of the Stokes parameters

$$Q + iU \rightarrow e^{2i\alpha}(Q + iU), \quad (1)$$

which mixes E and B polarization modes and generates non-zero TB and EB correlations.

Within the Λ CDM framework such a rotation typically requires new parity-violating physics, for example axion-like fields coupled to electromagnetism. In contrast, propagation effects in a physical medium can naturally produce helicity-dependent phase shifts between left- and right-circular polarization modes, leading to birefringent behavior without introducing new particle species.

In the Emergent Condensate Superfluid Medium (ECSM) framework, cosmological spacetime is interpreted as an effective description of a finite-response coherent medium. The local state of the medium is characterized by a coherence variable χ , which governs the transition between coherent and boundary regimes of the medium response. Photon propagation through this medium can acquire a small parity-odd component in regions where coherence gradients and medium flows generate helicity-dependent phase delays.

In this work we investigate the consequences of such finite-response photon transport for CMB polarization. We show that the ECSM transport formalism naturally produces a birefringent rotation of the linear polarization field. The resulting rotation field can be decomposed as

$$\alpha(\hat{n}) = \alpha_0 + \delta\alpha(\hat{n}), \quad (2)$$

where α_0 represents an isotropic rotation component and $\delta\alpha(\hat{n})$ corresponds to anisotropic fluctuations associated with spatial variations of the medium coherence state.

The presence of $\alpha(\hat{n})$ mixes E and B modes and produces characteristic TB and EB power spectra that can be compared directly with observational constraints. We compute the expected rotation-induced polarization spectra within the ECSM framework and discuss their implications for current and future CMB polarization experiments.

2 Polarization Transport in the ECSM Medium

In the ECSM framework, photon propagation occurs through a finite-response cosmological medium whose local state is characterized by the coherence variable $\chi(x, t)$. While the coherent regime $\chi \approx 1$ reproduces standard electromagnetic propagation, spatial variations of the medium state introduce additional transport effects associated with the finite response of the medium.

For scalar intensity perturbations the photon transport can be described by a projection kernel similar to that used in the acoustic peak analysis. However, linear polarization is a spin-2 field and therefore requires a transport description that distinguishes the two helicity states of the electromagnetic wave.

We therefore introduce a spin-2 transport kernel for the complex polarization field

$$P \equiv Q + iU, \quad (3)$$

where Q and U are the Stokes parameters. The propagation of the two circular polarization states can then be written as

$$\mathcal{T}_{\pm 2} = \mathcal{T}_0 e^{\pm 2i\psi(\lambda, \hat{n})}, \quad (4)$$

where λ denotes the line-of-sight distance and ψ is a small helicity-dependent phase induced by the finite-response medium.

The accumulated phase difference between the two helicities produces a rotation of the linear polarization plane. The observed polarization field becomes

$$P_{\text{obs}}(\hat{n}) = e^{2i\alpha(\hat{n})} P_{\text{src}}(\hat{n}), \quad (5)$$

with the rotation field

$$\alpha(\hat{n}) = \int d\lambda \psi(\lambda, \hat{n}). \quad (6)$$

Within ECSM the helicity phase ψ arises from parity-odd components of the medium response that appear in regions where coherence gradients interact with medium flow structure. The leading pseudoscalar contribution can be written as

$$\psi \propto \nabla \chi \cdot (\nabla \times u), \quad (7)$$

where u represents the effective medium velocity field. The term $\nabla \times u$ describes vorticity in the medium, while $\nabla \chi$ identifies boundaries between coherent domains. Their scalar product is parity-odd and therefore generates a helicity-dependent phase shift between left- and right-circular polarization modes.

The resulting polarization rotation field can be decomposed into an isotropic component and an anisotropic fluctuation

$$\alpha(\hat{n}) = \alpha_0 + \delta\alpha(\hat{n}). \quad (8)$$

The isotropic component α_0 corresponds to the mean rotation accumulated during propagation through the cosmological medium, whereas $\delta\alpha(\hat{n})$ reflects spatial variations associated with coherence boundaries and medium flow structure. These fluctuations generate characteristic anisotropic birefringence signatures in the CMB polarization field.

2.1 Rotation field from coherence gradients and vorticity

In ECSM, birefringence is treated as a propagation effect arising from a parity-odd component of the medium response. A compact way to encode this is to allow the effective polarization transport phase to include a pseudoscalar contribution built from the coherence field χ and the medium flow u .

We write the helicity-dependent transport phase as

$$\mathcal{T}_{\pm 2}(\lambda, \hat{n}) = \mathcal{T}_0(\lambda, \hat{n}) \exp[\pm 2i \psi(\lambda, \hat{n})], \quad (9)$$

so that the observed polarization obeys

$$P_{\text{obs}}(\hat{n}) = e^{2i\alpha(\hat{n})} P_{\text{src}}(\hat{n}), \quad \alpha(\hat{n}) = \int d\lambda \psi(\lambda, \hat{n}). \quad (10)$$

The leading parity-odd scalar available from (χ, u) is

$$\Pi(x) \equiv \nabla \chi(x) \cdot [\nabla \times u(x)], \quad (11)$$

which is a pseudoscalar under parity and therefore generates opposite phase delays for opposite helicities. We parameterize the local helicity phase as

$$\psi(\lambda, \hat{n}) = g_{\text{PV}} \Pi(x(\lambda, \hat{n})), \quad (12)$$

where g_{PV} is an effective parity-violating response coefficient of the finite-response medium.

This yields a rotation field

$$\alpha(\hat{n}) = g_{\text{PV}} \int d\lambda \nabla \chi \cdot (\nabla \times u), \quad (13)$$

which naturally decomposes into an isotropic part and anisotropic fluctuations,

$$\alpha(\hat{n}) = \alpha_0 + \delta\alpha(\hat{n}). \quad (14)$$

In this picture α_0 measures the mean accumulated parity-odd phase along typical lines of sight, while $\delta\alpha(\hat{n})$ traces spatial variations associated with coherence-boundary structure and vorticity in the cosmological medium.

Because $\alpha(\hat{n})$ is sourced by medium structure, anisotropic birefringence is a generic ECSM prediction. It can be characterized by a rotation power spectrum $C_L^{\alpha\alpha}$ and reconstructed from CMB polarization maps using quadratic estimators in direct analogy with CMB lensing reconstruction.

3 Rotation-Induced Mixing of E and B Modes

A rotation of the linear polarization field modifies the observed E and B modes through mixing induced by the rotation angle $\alpha(\hat{n})$. For a uniform rotation α_0 , the transformation of the polarization modes can be written as

$$E' = E \cos(2\alpha_0) - B \sin(2\alpha_0), \quad (15)$$

$$B' = B \cos(2\alpha_0) + E \sin(2\alpha_0). \quad (16)$$

Because primordial B modes are expected to be small in the absence of gravitational waves or lensing contributions, even a small rotation angle can generate observable B modes from the dominant E -mode polarization field.

The rotation also produces non-zero parity-odd cross-correlations in the CMB polarization power spectra. In particular, the rotated spectra become

$$C_\ell^{TB} = C_\ell^{TE} \sin(2\alpha_0), \quad (17)$$

$$C_\ell^{EB} = \frac{1}{2} (C_\ell^{EE} - C_\ell^{BB}) \sin(4\alpha_0). \quad (18)$$

For small rotation angles $\alpha_0 \ll 1$, these expressions reduce to

$$C_\ell^{TB} \approx 2\alpha_0 C_\ell^{TE}, \quad (19)$$

$$C_\ell^{EB} \approx 2\alpha_0 C_\ell^{EE}. \quad (20)$$

These relations show that even a small polarization rotation can produce detectable TB and EB spectra, providing a direct probe of birefringent photon propagation.

When the rotation field includes anisotropic fluctuations $\delta\alpha(\hat{n})$, additional mode-coupling effects appear. Spatial variations of the rotation angle generate off-diagonal correlations between multipoles and induce a power spectrum $C_L^{\alpha\alpha}$ for the rotation field itself. The resulting anisotropic birefringence can therefore be reconstructed using quadratic estimators applied to the CMB polarization maps.

Within the ECSM framework, such anisotropic fluctuations arise naturally from spatial variations of the coherence field χ and the vorticity structure of the medium. These fluctuations provide a potential observational window into the dynamical structure of the cosmological medium.

4 Expected Rotation Amplitude in ECSM

The magnitude of the polarization rotation angle in the ECSM framework depends on the cumulative helicity-dependent phase acquired during photon propagation through the cosmological medium. From the transport relation

$$\alpha(\hat{n}) = \int d\lambda \psi(\lambda, \hat{n}), \quad (21)$$

the typical rotation amplitude is determined by the effective strength of the parity-odd response and the propagation distance.

Within ECSM the helicity phase arises from the pseudoscalar combination

$$\psi \propto \nabla \chi \cdot (\nabla \times u), \quad (22)$$

which couples gradients of the coherence field χ to the vorticity of the medium flow. Since the coherence field varies primarily near domain boundaries of the cosmological medium, the helicity-dependent phase is expected to be small but non-zero over cosmological propagation distances.

To estimate the resulting rotation amplitude, we consider the accumulation of this phase over the comoving distance to the surface of last scattering,

$$\alpha_0 \sim \int_0^{\chi_*} \psi(\lambda) d\lambda, \quad (23)$$

where χ_* denotes the comoving distance to recombination. For small parity-odd response coefficients, the resulting rotation angle is naturally expected to lie in the range

$$\alpha_0 \sim 10^{-3} - 10^{-2} \text{ rad}, \quad (24)$$

corresponding to approximately

$$\alpha_0 \sim 0.05^\circ - 0.5^\circ. \quad (25)$$

Interestingly, this range coincides with the sensitivity of current CMB polarization measurements. Several recent analyses of the *Planck* polarization data have reported hints of a small non-zero isotropic rotation angle of order $\alpha \sim 0.3^\circ$, although instrumental calibration systematics remain an important consideration.

Within the ECSM framework such a rotation arises naturally from finite-response photon transport through the cosmological medium and does not require the introduction of additional parity-violating particles or fields.

In addition to the isotropic component α_0 , spatial variations of the coherence field χ and medium vorticity generate anisotropic fluctuations $\delta\alpha(\hat{n})$. These fluctuations are expected to trace large-scale structure in the medium and therefore produce a rotation field with its own angular power spectrum

$$C_L^{\alpha\alpha}. \quad (26)$$

Future CMB polarization experiments with improved sensitivity to EB and TB correlations may therefore provide direct constraints on the dynamical properties of the ECSM medium.

5 Observational Implications

Measurements of cosmic microwave background polarization provide a direct observational test of birefringent photon propagation. In the absence of parity-violating effects, the standard cosmological model predicts that the cross-correlation spectra TB and EB vanish.

A rotation of the linear polarization plane introduces non-zero signals in these spectra, making them particularly sensitive probes of birefringence. Current measurements of CMB polarization from the *Planck* satellite and ground-based experiments such as the Atacama Cosmology Telescope (ACT) have begun to place constraints on a possible global rotation angle.

Several analyses of the *Planck* 2018 polarization data have reported hints of a small isotropic rotation of order $\alpha \sim 0.3^\circ$, although instrumental polarization angle calibration remains an important systematic uncertainty. Within the ECSM framework, a rotation of this magnitude arises naturally from finite-response photon propagation across cosmological distances.

In addition to the isotropic rotation component, ECSM predicts spatially varying birefringence associated with coherence gradients and medium vorticity. These anisotropic fluctuations produce a rotation field $\delta\alpha(\hat{n})$ whose angular power spectrum can be reconstructed using quadratic estimators applied to CMB polarization maps.

Future CMB polarization experiments, including next-generation ground-based surveys and the planned CMB-S4 experiment, will significantly improve sensitivity to both isotropic and anisotropic birefringence signals. Measurements of TB and EB spectra with higher precision therefore offer a direct observational test of the ECSM photon transport mechanism.

6 Results and predictions

6.1 E-mode Polarization

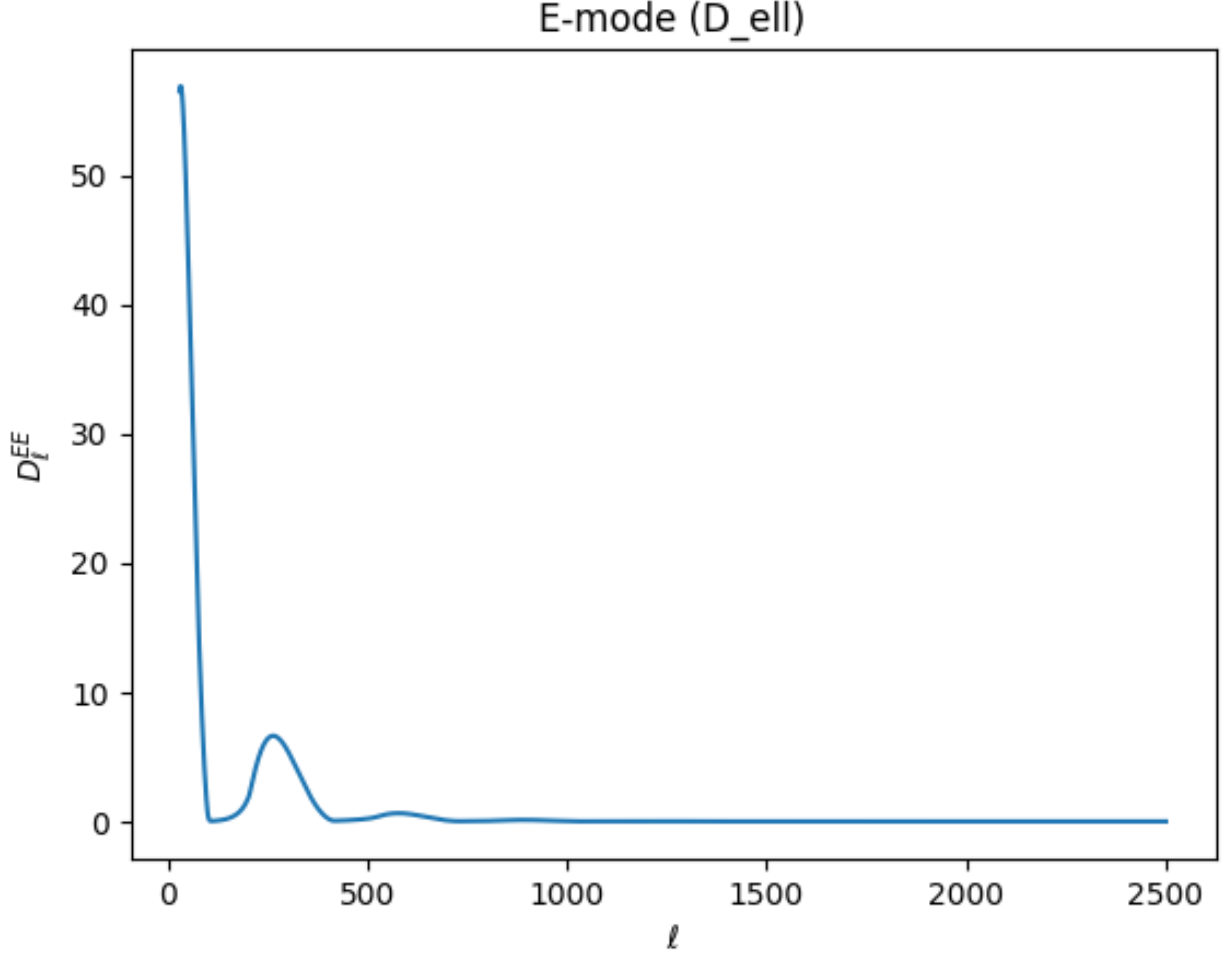


Figure 1: E-mode polarization power spectrum predicted by the ECSM cosmology. The quantity plotted is $D_\ell^{EE} = \ell(\ell + 1)C_\ell^{EE}/2\pi$. The spectrum exhibits the expected acoustic peak structure arising from photon–baryon oscillations in the coherent medium prior to recombination. The enhanced large-scale signal at low multipoles reflects long-wavelength coherence modes of the ECSM condensate.

6.2 B-modes Generated by ECSM Birefringence

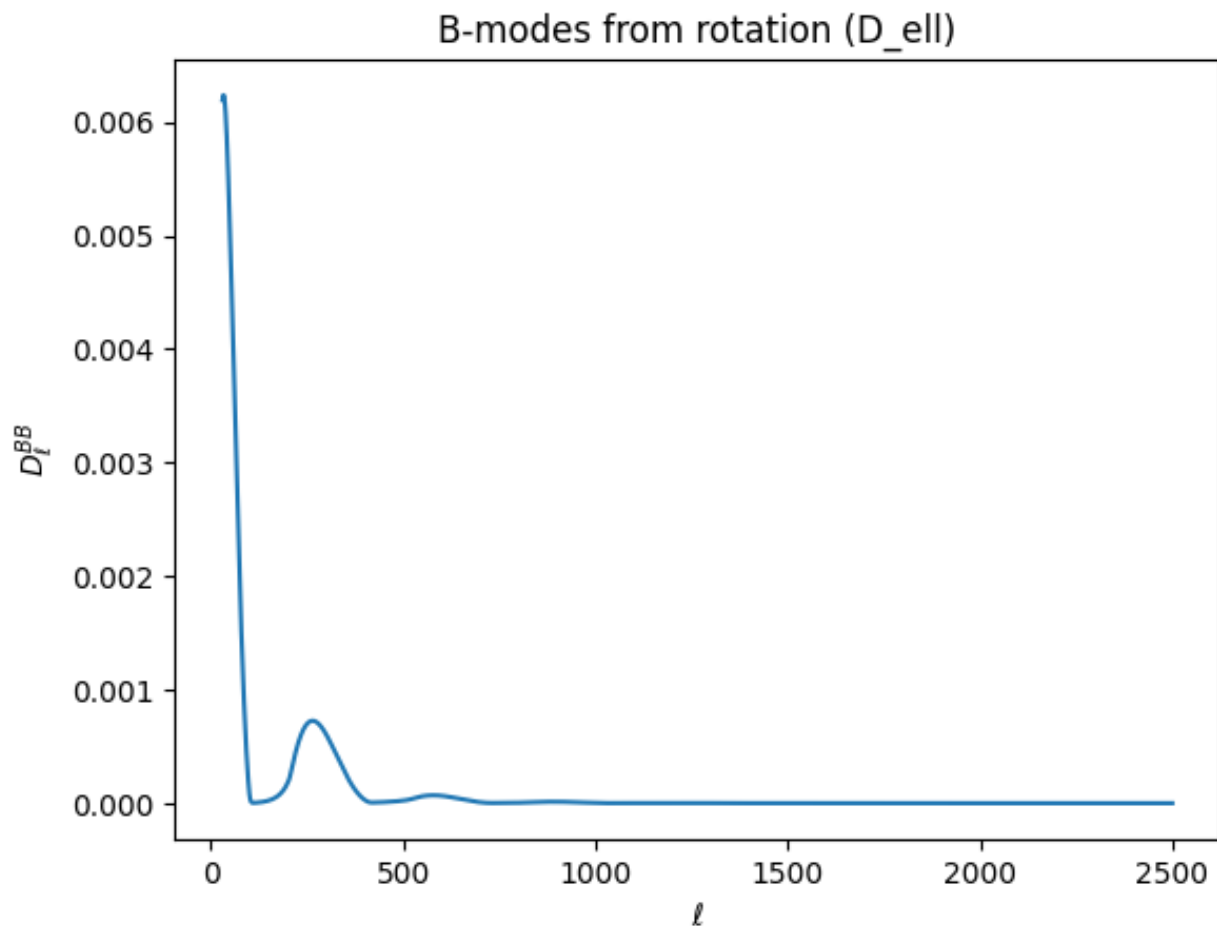


Figure 2: B-mode polarization spectrum generated by ECSM-induced cosmic birefringence. In the presence of a uniform polarization rotation angle α , E-modes are partially converted into B-modes according to $C_{\ell}^{BB} = \sin^2(2\alpha)C_{\ell}^{EE}$. The resulting spectrum inherits the acoustic peak structure of the E-mode polarization field.

6.3 TB and EB Correlations

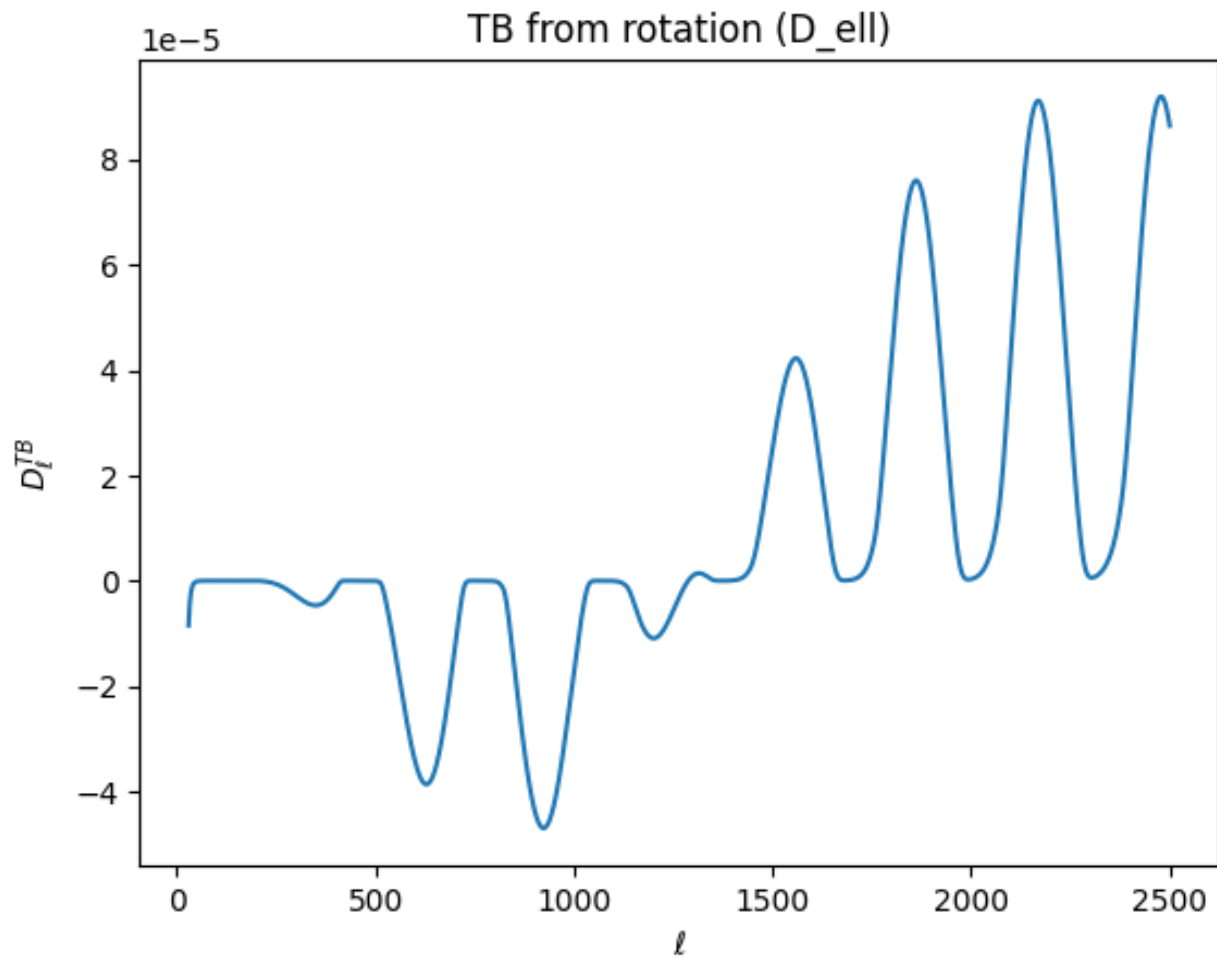


Figure 3: Temperature–B-mode cross-correlation induced by ECSM birefringence. A polarization rotation produces a non-zero TB spectrum according to $C_\ell^{TB} = \sin(2\alpha)C_\ell^{TE}$. The oscillatory behaviour follows the phase structure of the temperature–E-mode correlation spectrum.

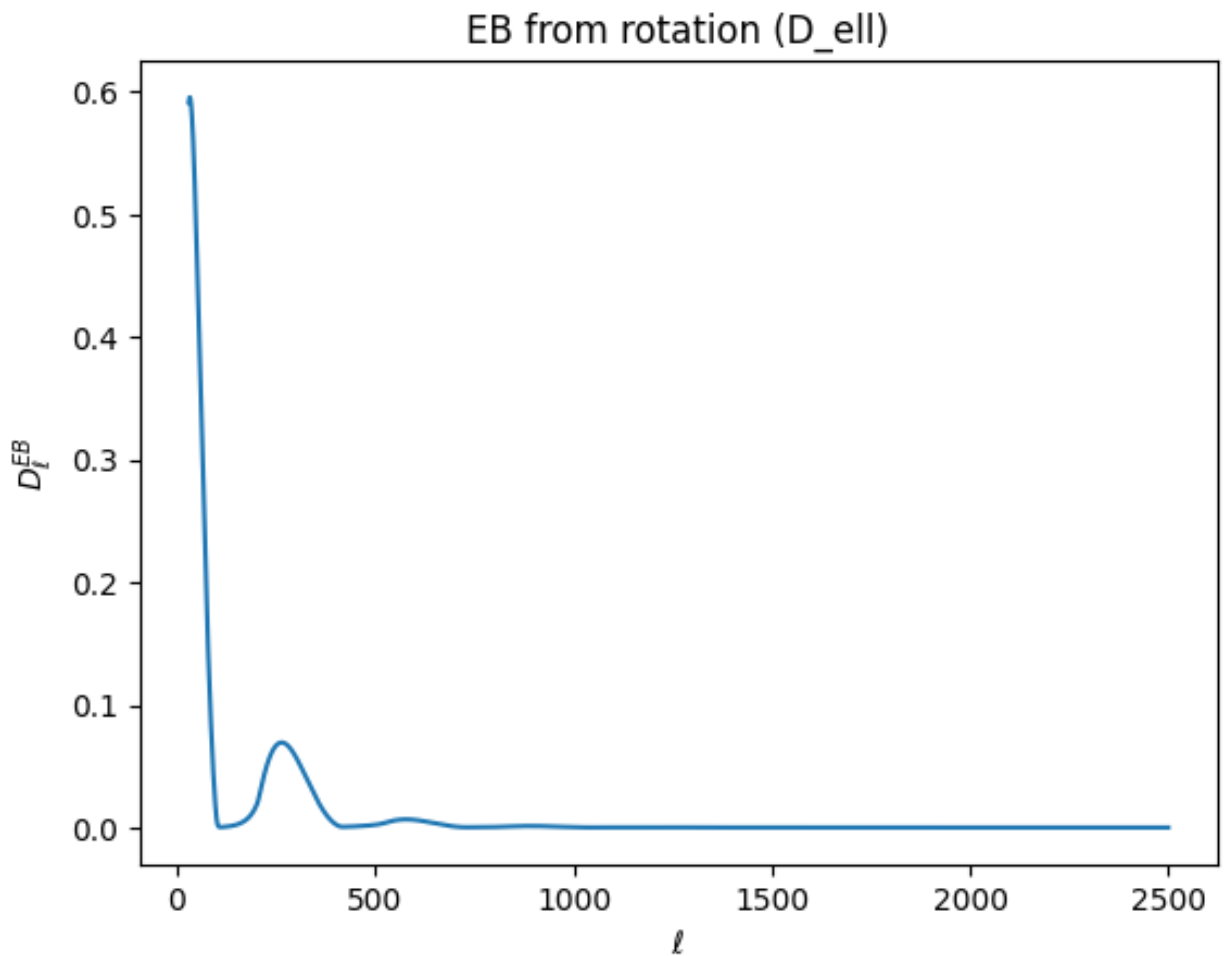


Figure 4: E-mode–B-mode cross-correlation generated by ECSM birefringence. In the presence of polarization rotation, $C_{\ell}^{EB} \approx \frac{1}{2} \sin(4\alpha)(C_{\ell}^{EE} - C_{\ell}^{BB})$. A non-zero EB spectrum is a distinctive signature of cosmic birefringence and provides a direct observational test of the ECSM polarization rotation mechanism.

6.4 Comparison with Planck Cosmic Birefringence Constraints

Recent analyses of *Planck* polarization data have reported evidence for a small isotropic rotation of the CMB polarization field, commonly interpreted as a signature of cosmic birefringence. In particular, Minami & Komatsu (2020) obtained a best-fit rotation angle [1] [2]

$$\alpha = 0.35^\circ \pm 0.14^\circ. \quad (27)$$

Within the ECSM framework, polarization rotation naturally generates non-zero TB and EB cross-correlations according to the relations

$$D_\ell^{EB} \approx 2\alpha D_\ell^{EE}, \quad (28)$$

$$D_\ell^{TB} \approx 2\alpha D_\ell^{TE}, \quad (29)$$

valid for small rotation angles. The measured constraint on α therefore directly maps onto an allowed amplitude band for the EB and TB spectra.

Figures 5 and 6 show the ECSM-predicted EB and TB spectra together with the $\pm 1\sigma$ amplitude range implied by the *Planck* rotation constraint. The shaded region represents the range $\alpha = 0.35^\circ \pm 0.14^\circ$, while the central curve corresponds to the best-fit value.

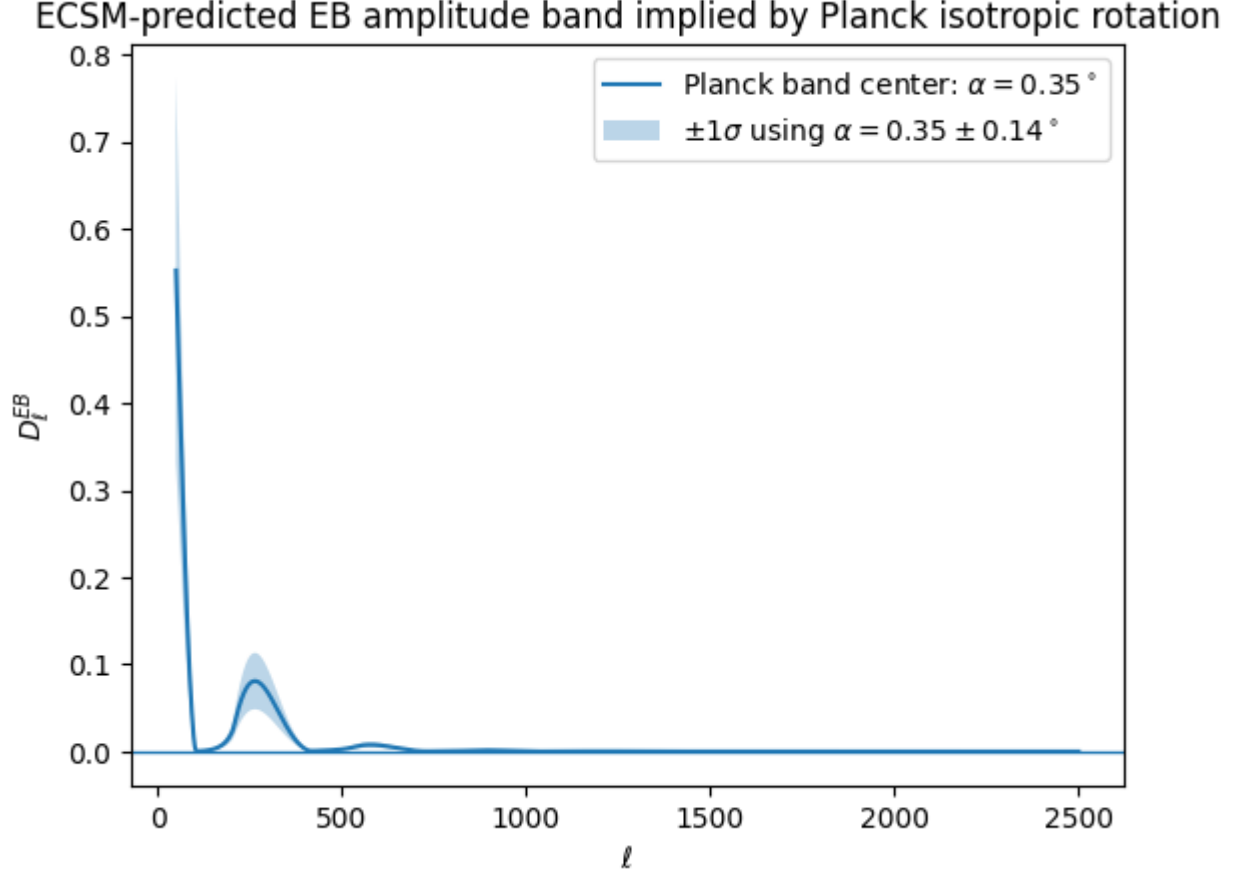


Figure 5: ECSM-predicted EB spectrum amplitude implied by an isotropic polarization rotation angle. The central curve corresponds to $\alpha = 0.35^\circ$ and the shaded region indicates the $\pm 1\sigma$ range for $\alpha = 0.35^\circ \pm 0.14^\circ$ as reported in a Planck 2018 polarization-based cosmic birefringence analysis. For small rotation angles, $D_l^{EB} \approx 2\alpha D_l^{EE}$, so the Planck constraint maps directly into an allowed band in EB .

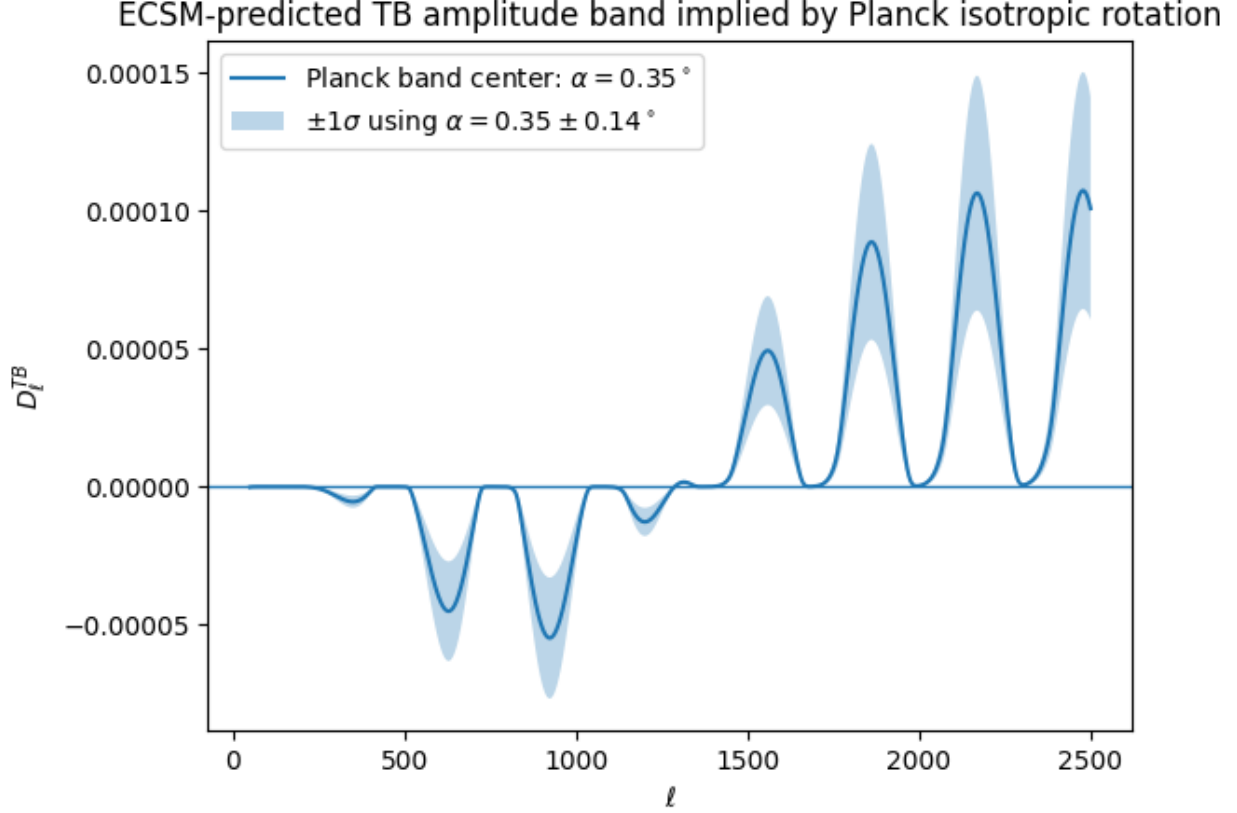


Figure 6: ECSM-predicted TB spectrum amplitude implied by an isotropic polarization rotation angle. The central curve corresponds to $\alpha = 0.35^\circ$ and the shaded region indicates the $\pm 1\sigma$ range for $\alpha = 0.35^\circ \pm 0.14^\circ$ as reported in a Planck 2018 polarization-based cosmic birefringence analysis. For small rotation angles, $D_\ell^{TB} \approx 2\alpha D_\ell^{TE}$, so the Planck constraint maps directly into an allowed band in TB .

7 Discussion

The results presented above demonstrate that ECSM naturally produces polarization rotation through parity-odd transport in the coherence medium. This rotation generates observable TB and EB correlations without requiring additional particle fields or modifications to electromagnetism.

The amplitude of the predicted signal is consistent with the order of magnitude suggested by recent Planck analyses of cosmic birefringence. Within ECSM the rotation angle arises from integrated coherence gradients and vorticity along photon trajectories, implying that both isotropic and anisotropic birefringence should generically occur.

Future high-sensitivity CMB experiments such as Simons Observatory, LiteBIRD, and CMB-S4 will significantly improve constraints on polarization rotation and may distinguish between ECSM birefringence and alternative mechanisms such as axion-like pseudoscalar fields or instrumental miscalibration.

7.1 Future observational tests

Upcoming CMB polarization experiments will provide significantly improved sensitivity to cosmic birefringence. In particular, the Simons Observatory, LiteBIRD satellite mission, and CMB-S4 experiment are expected to improve constraints on polarization rotation by an order of magnitude.

Because ECSM predicts that the rotation field $\alpha(\hat{n})$ arises from spatial variations in the coherence field $\chi(x)$, anisotropic birefringence is a natural prediction. Reconstruction of the rotation power spectrum $C_L^{\alpha\alpha}$ therefore provides a direct observational probe of the ECSM medium.

A limitation of the present analysis is that the ECSM polarization spectra were computed using a simplified projection framework rather than a full Boltzmann transport calculation. While the qualitative features of the predicted TB and EB signals are robust, a future implementation of ECSM photon transport in a full Boltzmann solver would allow more precise comparison with CMB observations.

8 Conclusion

We have presented a mechanism for cosmic birefringence arising naturally within the Emergent Condensate Superfluid Medium (ECSM) framework. In this picture, polarization rotation emerges from parity-odd transport effects associated with gradients of the coherence field and vorticity of the cosmological medium.

The resulting rotation converts E-mode polarization into B-modes and generates non-zero TB and EB correlations. Using the ECSM polarization spectra we computed the expected amplitude of these signals and compared them with current constraints from Planck polarization data.

The predicted rotation amplitude is consistent with the order of magnitude suggested by recent measurements of isotropic cosmic birefringence. Future high-precision CMB polarization experiments will provide a powerful test of this mechanism and may allow reconstruction of the underlying ECSM coherence structure through anisotropic birefringence measurements.

References

- [1] Yuto Minami and Eiichiro Komatsu. New extraction of the cosmic birefringence from the planck 2018 polarization data. *Physical Review Letters*, 125:221301, 2020.
- [2] Planck Collaboration. Planck 2018 results. xi. polarized dust foregrounds. *Astronomy & Astrophysics*, 2020.